

Linear Algebra – MAT 2610

Section 2.8 (Subspaces of R^n)

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Consider this set:

$$S = \left\{ \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} \in R^3 : 2s_1 - s_2 + s_3 = 0 \right\}$$

This set is any vector $\begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix}$ that satisfies the condition $2s_1 - s_2 + s_3 = 0$

Examples of vectors in S include $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 3 \\ -1 \\ -7 \end{bmatrix}$ among many others.

$$2(3) - (-1) + (-7) = 6 + 1 - 7 = 0 \text{ so } \begin{bmatrix} 3 \\ -1 \\ -7 \end{bmatrix} \in S \quad \left| \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix} \notin S$$

Consider this set:

$$S = \left\{ \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} \in \mathbb{R}^3 : 2s_1 - s_2 + s_3 = 0 \right\}$$

Suppose that we have two vectors $u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$ and $v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$ that are both in S . Then

$$2u_1 - u_2 + u_3 = 0$$

$$2v_1 - v_2 + v_3 = 0$$

Now look at the vector $(u + v) = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \end{bmatrix}$. Then

$$2(u_1 + v_1) - (u_2 + v_2) + (u_3 + v_3) = [2u_1 - u_2 + u_3] + [2v_1 - v_2 + v_3] = 0 + 0 = 0$$

This means that the vector $u + v$ is also in S .



Consider this set:

$$S = \left\{ \begin{bmatrix} s_1 \\ s_2 \\ s_3 \end{bmatrix} \in \mathbb{R}^3 : 2s_1 - s_2 + s_3 = 0 \right\}$$

Now suppose $u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$ is in S and k is any scalar. Then

$$2u_1 - u_2 + u_3 = 0$$

Now look at the vector $ku = \begin{bmatrix} ku_1 \\ ku_2 \\ ku_3 \end{bmatrix}$. Then

$$2(ku_1) - (ku_2) + (ku_3) = k(2u_1 - u_2 + u_3) = k\underline{0} = \underline{0}$$

This means that the vector ku is also in S .



Definition: A set W is a **subspace** of R^n if it satisfies the following three properties:

1. The zero vector is in W
2. Whenever u and v are in W , then so is $u + v$ (closed under vector addition)
3. Whenever u is in W and c is a scalar, then cu is also in W (closed under scalar multiplication)

Notice that our earlier example S is a subspace of R^3



Example: Is the set $S = \left\{ \begin{bmatrix} s_1 \\ s_2 \\ 0 \\ 0 \end{bmatrix} \in R^4 : \underline{s_1, s_2 \text{ scalars}} \right\}$ a subspace of R^4 ?

(1) $0 \in S$? Let $s_1 = 0, s_2 = 0$, then $v = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \in S$ ✓

(2) Let $u = \begin{bmatrix} u_1 \\ u_2 \\ 0 \\ 0 \end{bmatrix}$ $v = \begin{bmatrix} v_1 \\ v_2 \\ 0 \\ 0 \end{bmatrix}$. Then $u + v = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ 0 \\ 0 \end{bmatrix} \in S$ ✓

(3) Let $u = \begin{bmatrix} u_1 \\ u_2 \\ 0 \\ 0 \end{bmatrix}$ k scalar. Then $k u = \begin{bmatrix} k u_1 \\ k u_2 \\ 0 \\ 0 \end{bmatrix} \in S$ ✓

So S is a subspace



Example: Is the set $S = \left\{ \begin{bmatrix} s_1 \\ s_2 \\ 0 \\ 0 \end{bmatrix} \in \mathbb{R}^4 : s_1, s_2 \geq 0 \right\}$ a subspace of $\mathbb{R}^4$?

Not a
Subspace

(1) $0 \in S : \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \in S$ ✓

(2) $u = \begin{bmatrix} u_1 \\ u_2 \\ 0 \\ 0 \end{bmatrix} \quad v = \begin{bmatrix} v_1 \\ v_2 \\ 0 \\ 0 \end{bmatrix}$ Then $u+v = \begin{bmatrix} u_1+v_1 \\ u_2+v_2 \\ 0 \\ 0 \end{bmatrix}$ ✓

Since $u_1, v_1 \geq 0$ then $u_1+v_1 \geq 0$
Since $u_2, v_2 \geq 0$ then $u_2+v_2 \geq 0$

(3) $u = \begin{bmatrix} u_1 \\ u_2 \\ 0 \\ 0 \end{bmatrix} \quad k = -1$. Then $ku = \begin{bmatrix} k u_1 \\ k u_2 \\ 0 \\ 0 \end{bmatrix}$ ✗



Consider the set R^n

1. R^n is a subspace of itself
2. $\{0\}$ is a subspace of R^n {Note: this is called the **zero subspace** of R^n }

$$\{0\} = \left\{ \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \right\}$$

$$(1) \quad 0 \in S \quad \checkmark$$

$$(2) \quad \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad \checkmark$$

$$(3) \quad k \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad \checkmark$$



Example: Let $S = \text{Span} \left\{ \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \right\}$. Show that S is a subspace of $\mathbb{R}^3$.

$$(1) \quad 0 \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \checkmark$$

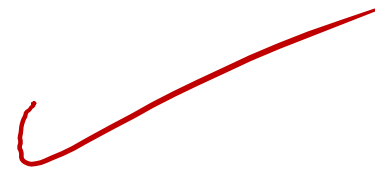
$$(2) \quad u = u_1 \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} + u_2 \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \quad v = v_1 \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} + v_2 \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

$$\begin{aligned} u + v &= \left[u_1 \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} + u_2 \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \right] + \left[v_1 \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} + v_2 \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \right] \\ &= (u_1 + v_1) \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} + (u_2 + v_2) \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \quad \checkmark \end{aligned}$$



$$(3) \quad u = u_1 \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} + u_2 \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \quad k \text{ is a scalar}$$

$$\begin{aligned} ku &= k \left[u_1 \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} + u_2 \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \right] \\ &= \underbrace{(ku_1)} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} + \underbrace{(ku_2)} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \end{aligned}$$



S is a subspace



The **null space** of a matrix A is the set of solutions to $Ax = 0$, denoted $Null(A)$

Example: Show the null space of $A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 3 \end{bmatrix}$ is a subspace of $\mathbb{R}^3$.

$$\begin{aligned} & \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ & \begin{bmatrix} 2 & 1 & -1 & 0 \\ 1 & 2 & 3 & 0 \end{bmatrix} \xrightarrow{r_1 - 2r_2 \rightarrow r_1} \begin{bmatrix} 2 & 1 & -1 & 0 \\ 0 & -3 & -7 & 0 \end{bmatrix} \xrightarrow{3r_1 + r_2 \rightarrow r_1} \\ & \begin{bmatrix} 6 & 0 & -10 & 0 \\ 0 & -3 & -7 & 0 \end{bmatrix} \xrightarrow{\begin{matrix} \frac{1}{6}r_1 \rightarrow r_1 \\ -\frac{1}{3}r_2 \rightarrow r_2 \end{matrix}} \begin{bmatrix} 1 & 0 & -\frac{4}{3} & 0 \\ 0 & 1 & \frac{7}{3} & 0 \end{bmatrix} \end{aligned}$$



$$\begin{bmatrix} x_1 & x_2 & -4/3 x_3 & 0 \\ 0 & 1 & 7/3 & 0 \end{bmatrix} \quad \begin{aligned} x_1 - 4/3 x_3 &= 0 \\ x_2 + 7/3 x_3 &= 0 \end{aligned}$$

$$\begin{aligned} x_1 &= \frac{4}{3} x_3 \\ x_2 &= -\frac{7}{3} x_3 \\ x_3 &= x_3 \end{aligned} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 4/3 \\ -7/3 \\ 1 \end{bmatrix} \leftarrow$$

$$\Rightarrow \text{Null}(A) = \text{Span} \left\{ \begin{bmatrix} 4/3 \\ -7/3 \\ 1 \end{bmatrix} \right\}$$

So $\text{Null}(A)$ is a subspace



Definition: The **column space** of a matrix A is the span of its columns, denoted $\text{col}(A)$

Example: Show the column space of $A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 3 \end{bmatrix}$ is $\mathbb{R}^2$.

$$S = \text{Span} \left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \end{bmatrix} \right\}$$

$$x_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \quad \text{for any } u_1, u_2 \text{ scalars}$$

$$\begin{bmatrix} 2 & 1 & -1 & u_1 \\ 1 & 2 & 3 & u_2 \end{bmatrix} \xrightarrow{r_1 - 2r_2 \rightarrow r_1} \begin{bmatrix} 2 & 1 & -1 & u_1 \\ 0 & -3 & -7 & u_1 - 2u_2 \end{bmatrix} \xrightarrow{3r_1 + r_2 \rightarrow r_1}$$

$$\begin{bmatrix} \boxed{6} & 0 & -10 & 4u_1 - 2u_2 \\ 0 & \boxed{-3} & -7 & u_1 - 2u_2 \end{bmatrix}$$



Definition: Let V be a non-zero subspace of R^n . A **basis** for V is a linearly independent set in V that spans V .

Example: The set of standard vectors $\{e_1, e_2, \dots, e_n\}$ is a basis for R^n . It is called the **standard basis**.

For instance, $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ is the standard basis of R^3

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = u_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + u_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + u_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$



Example: Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 2 & 2 & 2 \end{bmatrix}$. We know $\left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \right\}$ is a generating set for the

column space. But these vectors are not linearly independent (why?)

When we row reduce A , we get $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. This row reduced matrix has leading entries

in columns 1 and 2. Therefore $\left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \right\}$ is a basis for $\text{col}(A)$



Theorem 13

The pivot columns of a matrix A form a basis for the column space of A



Example: Find a basis for $\text{col}(A)$ where $A = \begin{bmatrix} 3 & 6 & 8 \\ 2 & 4 & 10 \\ 1 & 2 & 4 \end{bmatrix}$.

$$\begin{bmatrix} 3 & 6 & 8 \\ 2 & 4 & 10 \\ 1 & 2 & 4 \end{bmatrix} \xrightarrow{r_1 - 3r_3 \rightarrow r_1} \begin{bmatrix} 3 & 6 & 8 \\ 2 & 4 & 10 \\ 0 & 0 & -4 \end{bmatrix} \xrightarrow{2r_1 - 3r_2 \rightarrow r_2} \begin{bmatrix} 3 & 6 & 8 \\ 0 & 0 & -14 \\ 0 & 0 & -4 \end{bmatrix}$$

$$\xrightarrow{4r_2 - 14r_3 \rightarrow r_2} \begin{bmatrix} 3 & 6 & 8 \\ 0 & 0 & -14 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{col}(A) = \text{Span} \left\{ \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 8 \\ 10 \\ 4 \end{bmatrix} \right\}$$

$$\text{Basis for } \text{col}(A) = \left\{ \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 8 \\ 10 \\ 4 \end{bmatrix} \right\}$$



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